

YUVRAJ GOVIND RAO

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SUMMARY

Front-end engineer (B.Tech CS, 2026) with hands-on React/TypeScript experience building complex interactive UIs, now focused on accessibility engineering. Familiar with WCAG 2.x (Level AA) and WAI-ARIA, with direct screen-reader testing experience using NVDA. Background in structured technical evaluation and documentation from a formal consumer product trial program, translating defects into clear, prioritized reports that product teams acted on.

CORE SKILLS

Accessibility: WCAG 2.x (Level AA), WAI-ARIA Roles/States/Properties, Semantic HTML, Screen-Reader Testing (NVDA)

Front-End: HTML, CSS, JavaScript, React, TypeScript

Backend & APIs: Python, FastAPI, REST API Design, Microservices, AsyncIO

Testing & Documentation: Structured Bug/UX Documentation & Reporting, Pytest, GitHub Actions (CI)

Databases & Tools: PostgreSQL, SQLite, Git/GitHub, Docker / Docker Compose

EXPERIENCE

User Trial Program Participant — Nothing Technology — 2024 – 2025

- Surfaced **30+** usability, functional, and performance defects ahead of public launch, measured against structured severity and reproducibility criteria, by conducting systematic pre-release testing of unreleased consumer hardware and software under real-world usage conditions.
- Directly influenced **2** pre-launch product decisions by translating raw user friction into clear, prioritized written reports for product teams, using a standardized bug/UX documentation framework.

PROJECTS

macOS-Inspired Interactive Portfolio — [GitHub](#)

- Built a fully interactive, macOS-style desktop UI in React and TypeScript with draggable, resizable windows (Finder, Safari, Photos, Terminal) and a Zustand-based window manager handling open/close/focus/drag state across 5+ simulated apps.
- Implemented full dark-mode and iOS-style responsive behavior for mobile, plus a GSAP animation layer for interface transitions.
- Integrated a conversational “Ask AI” feature (Google Gemini) that answers visitor questions about background and projects directly within the interface.

GraphRAG — Knowledge Graph-Enhanced Q&A — [GitHub](#)

- Built Python/FastAPI microservices integrating Google Gemini and OpenAI APIs, combining Qdrant vector search with Neo4j graph traversal for cited, multi-document question answering.
- Engineered an automated entity/relationship extraction and resolution pipeline using UNWIND-batched Cypher writes to merge 100+ entities into Neo4j in under 1.5 seconds.

YouTubeRAG — Multi-Video RAG Assistant — [GitHub](#)

- Built a full-stack application (React/Vite, FastAPI, PostgreSQL + pgvector) that ingests YouTube playlists and answers cross-video questions with timestamp-grounded citations tied to an embedded video player.
- Implemented sentence-aware caption chunking enabling pgvector cosine similarity search over 768-dimension embeddings for clickable, second-accurate citations.
- Shipped two config-driven Docker Compose deployment profiles, with automated tests run via GitHub Actions CI.

EDUCATION

B.Tech, Computer Science & Engineering (Game Technology) — Galgotias University

2026 | CGPA: 7.0 / 10

CERTIFICATIONS

- Google Android Developer Certification